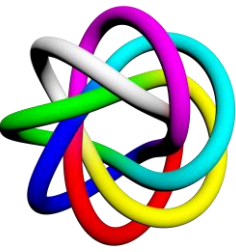


Modelling, Transformation and Software Generation

Simplified, InformationGrid and Mendix

Dr. ir. Mark A.T. Mulder

The
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Company 2



- 1990-1995 – Master Computer Science
- 1995-2017 – Various consultancy companies
Developer, Designer, Architect, Enterprise Engineer
- 2013-2022 – PhD on Automatic Verification and Exchange of DEMO
- 2017-Now – TEEC2 BV – 5 employees

- Volunteer

- Chairman Eei
- Miniworld Rotterdam

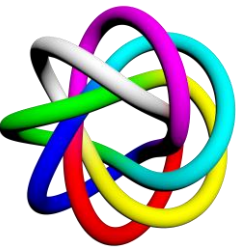


- Hobby

- Model Trains, Electronics, Motorcycles, Handyman

TEEC2: Our Services

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Modelling Software



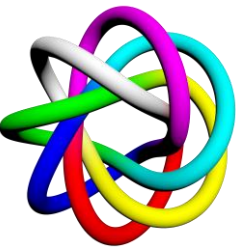
Education



Consulting

Problem statement

- Modelling in a small organisation is often skipped
- Modelling needs to bring something: RoME
- Repetitive (inconsistent) creation of operational structures



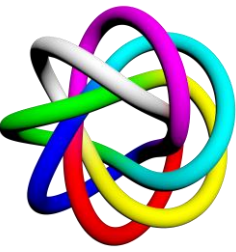
Solution perspective

Pros

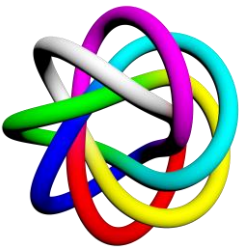
- Multiple usages of single model
- Lower Time-to-market
- Prototyping
- Consistency -> low maintenance effort

Cons

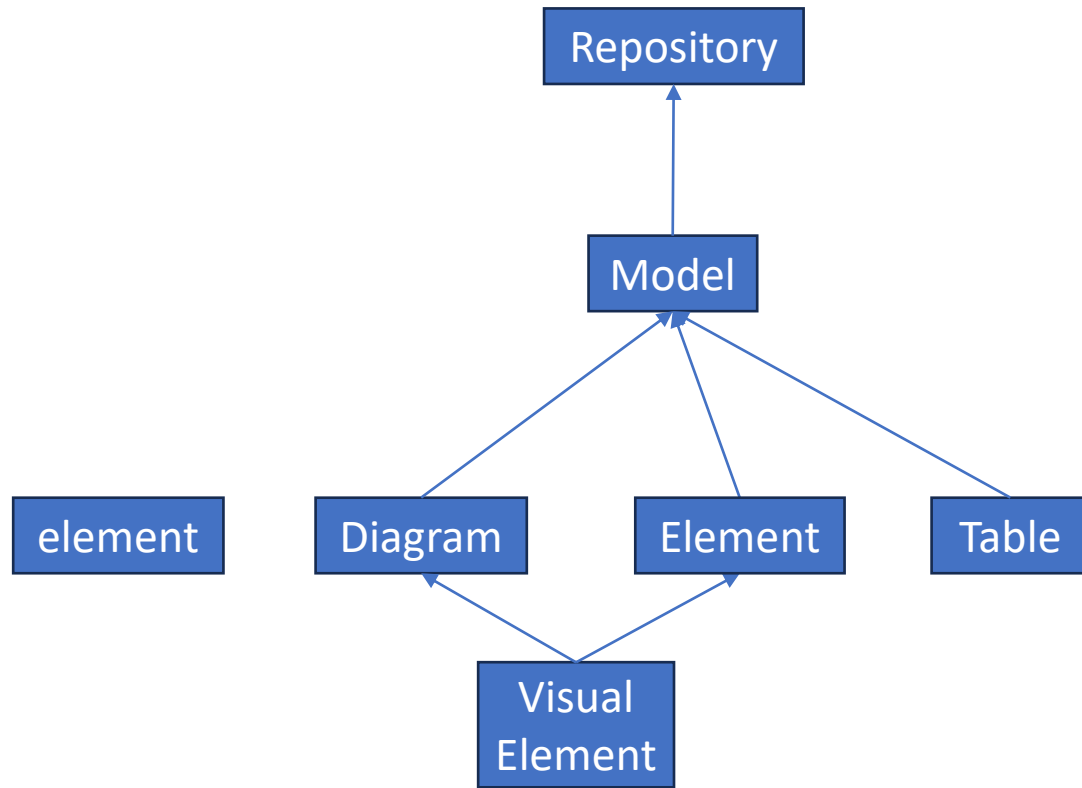
- Higher education
- Less flexible per concept
- Version management
- Data Migration in operation



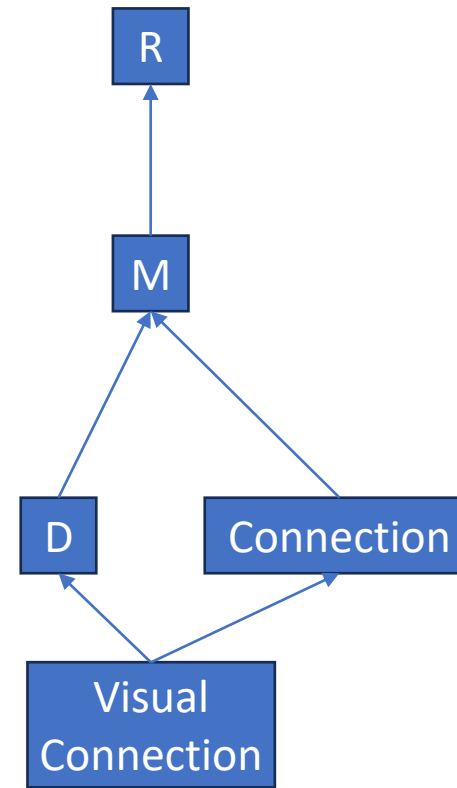
Theory: Component Hierarchy



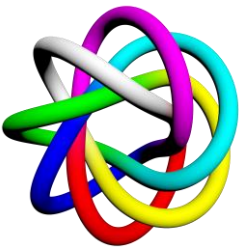
Elements



Connections

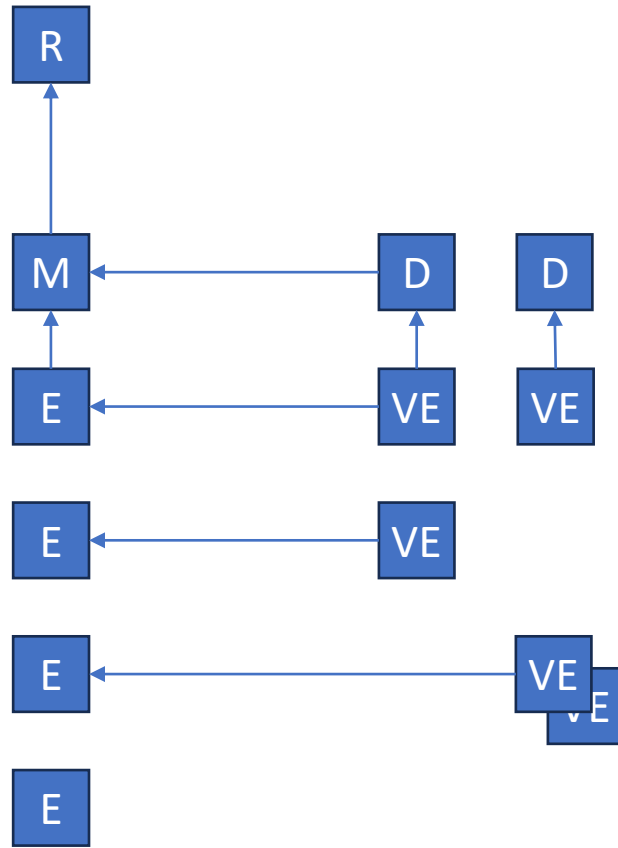
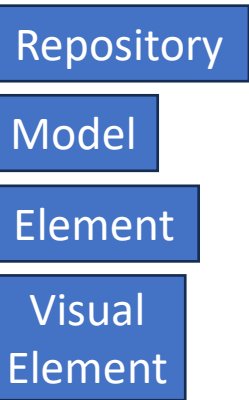


Theory: Models and Views



Model

View

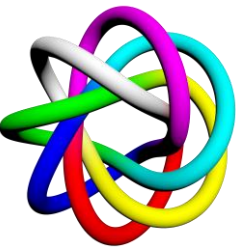


<- Same element on different diagrams

<- Same element multiple times on same diagram

<- Element on no diagram

Research on transformations



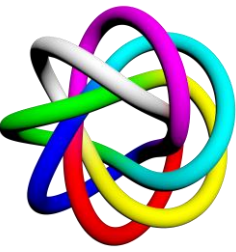
Master

- Towards a round-trip transformation between DEMO and VISI.
- (UT Wien) BPMN-SD transformation

PhD

- Describing the transformation language for graph-based models.
- Describing transformation rules for the evolvability of models.

Transformations – status



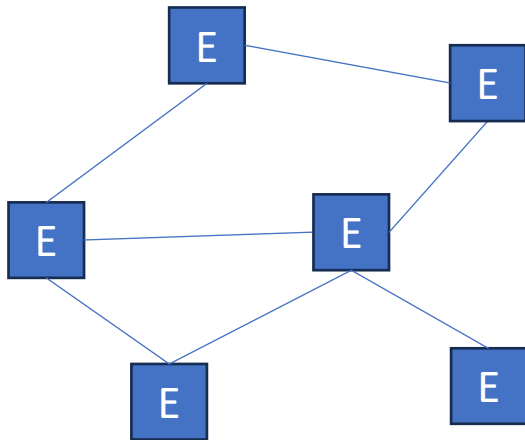
element



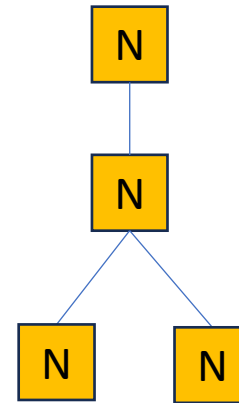
connection

Graph

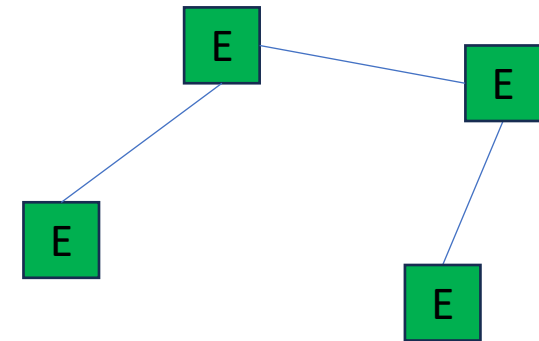
Source



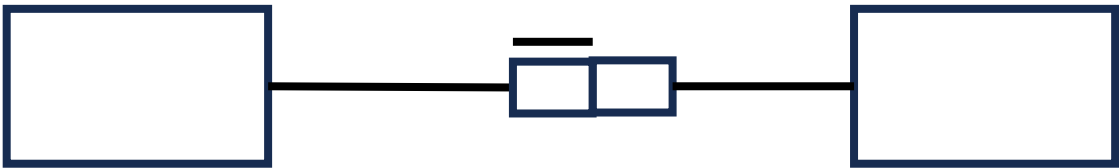
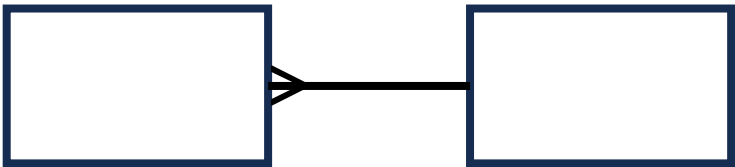
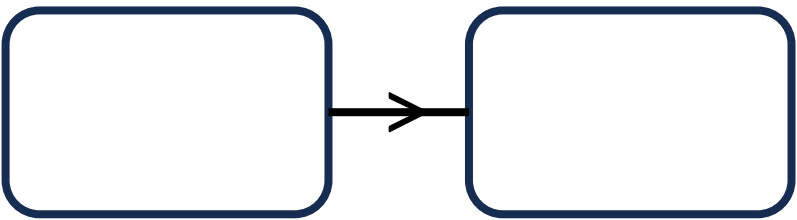
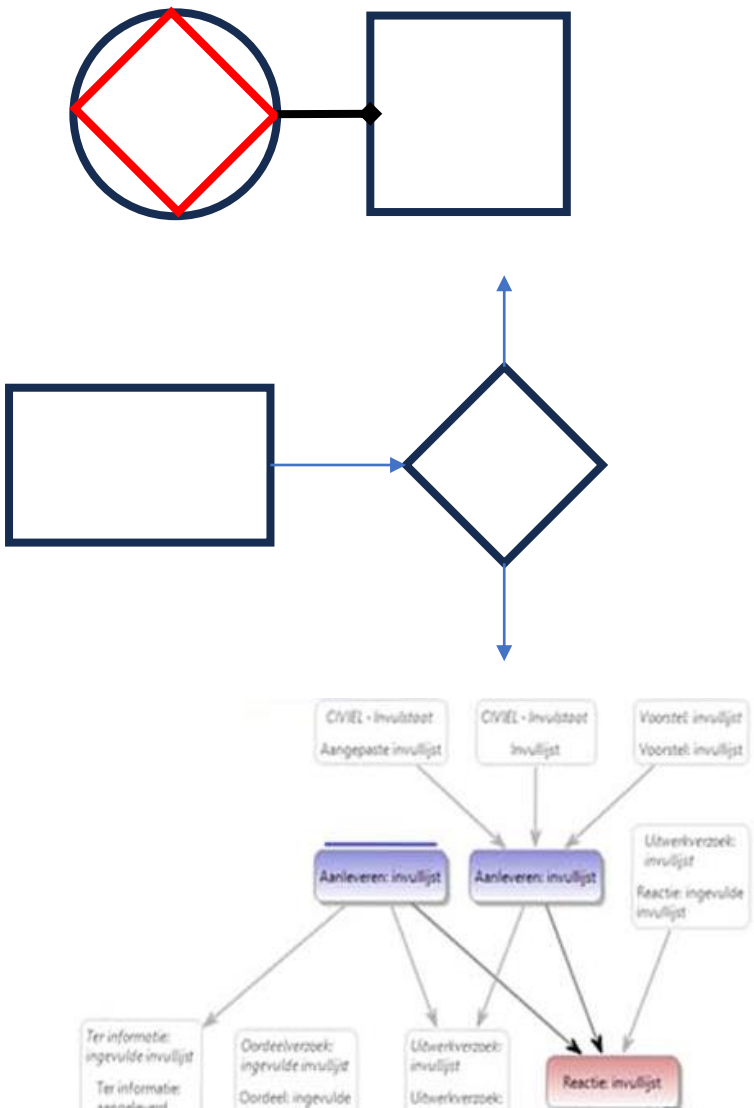
T-Tree



Target

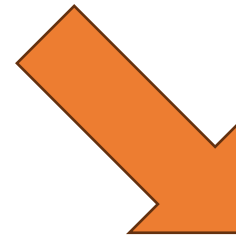
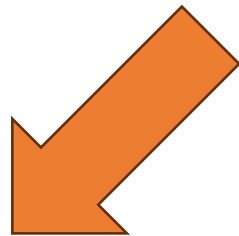
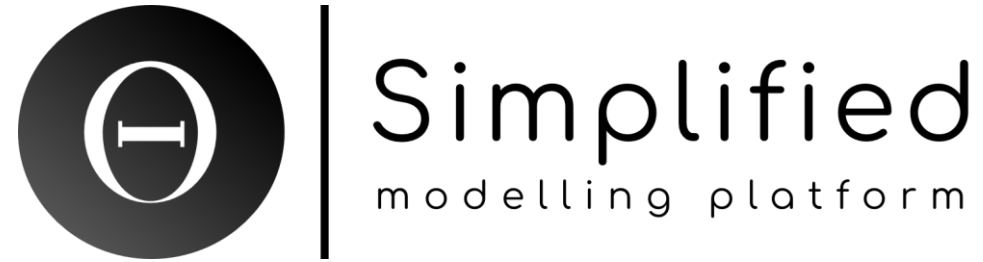
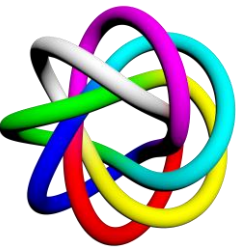


Process and Data



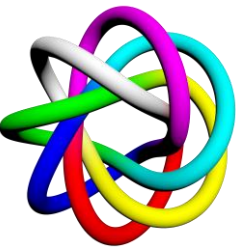
Generation

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Generatie opties

- Staging generation from API/JSON, CRUD
- Application Data generation, CRUD
- Application Data and Process generation

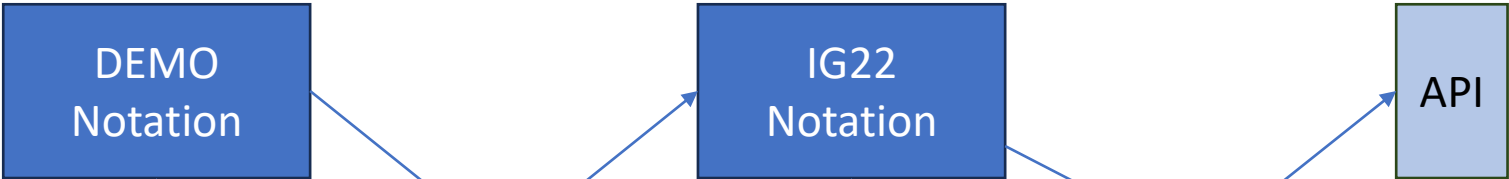


Architecture – Data, CRUD

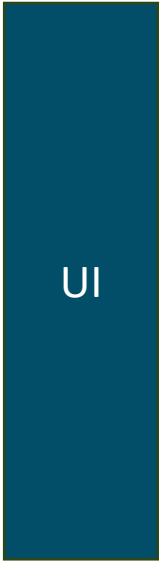
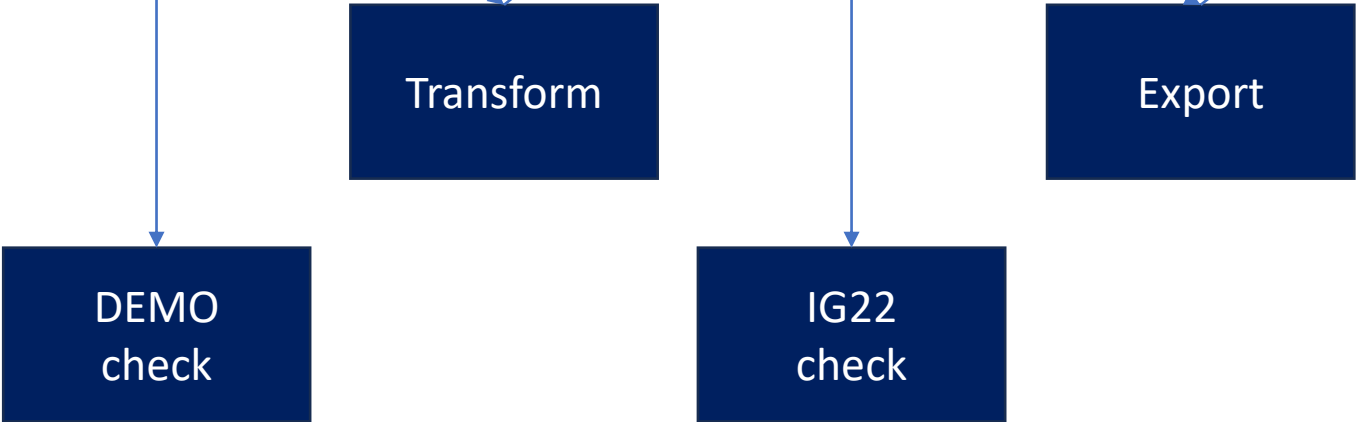
Applicaties



Notaties



Server
Extensies



Architecture – Data, Process

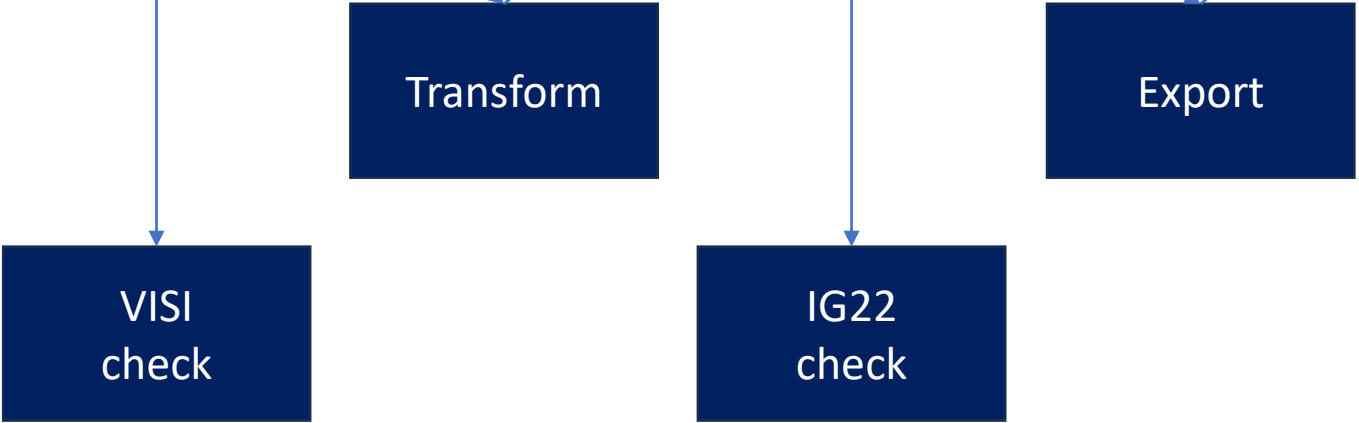
Applicaties



Notaties

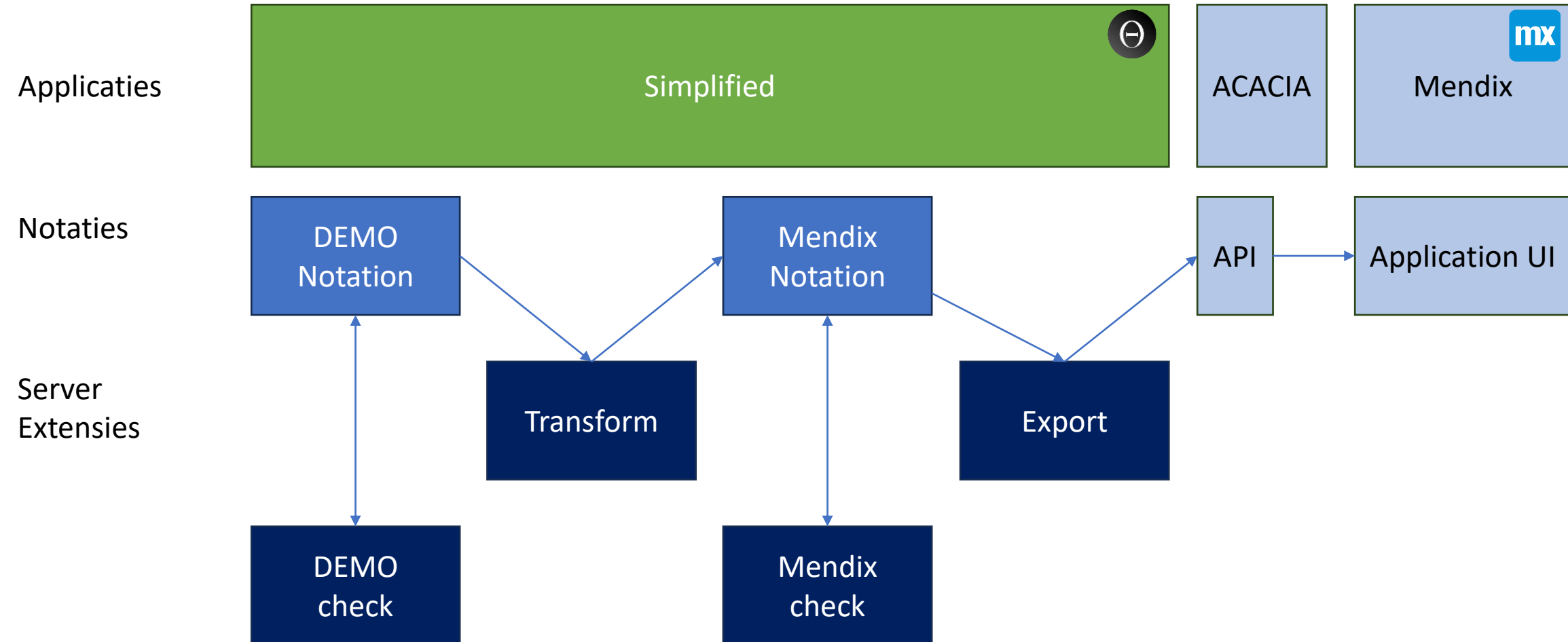
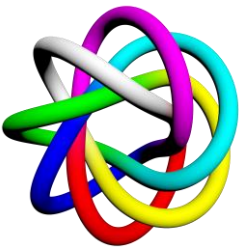


Server
Extensies



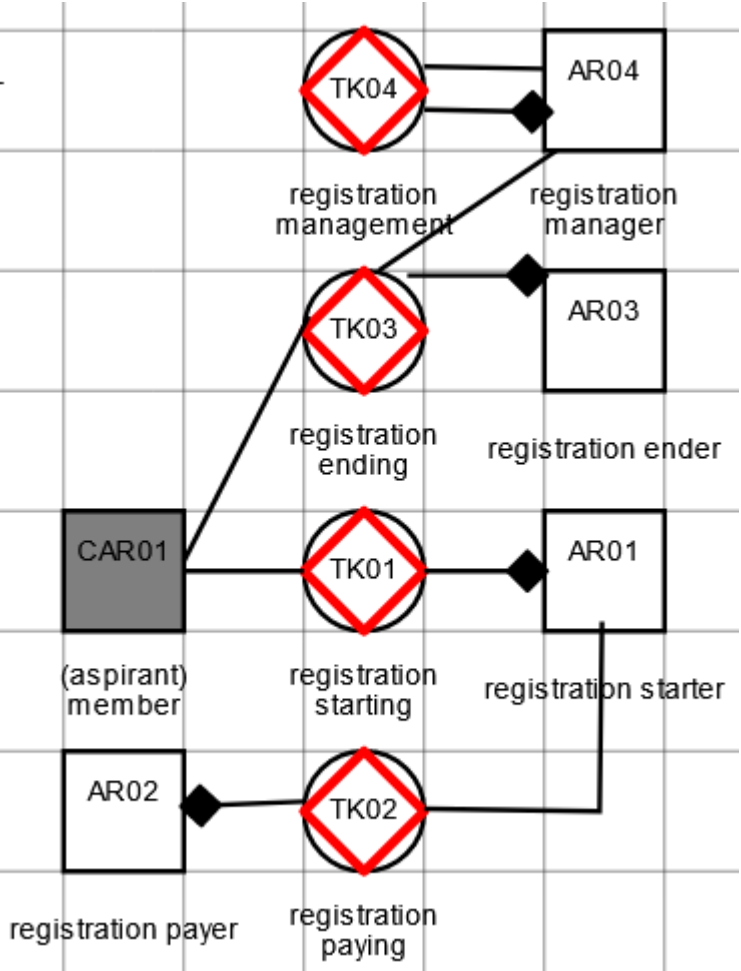
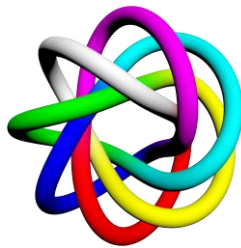
Current State – Mendix

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Example

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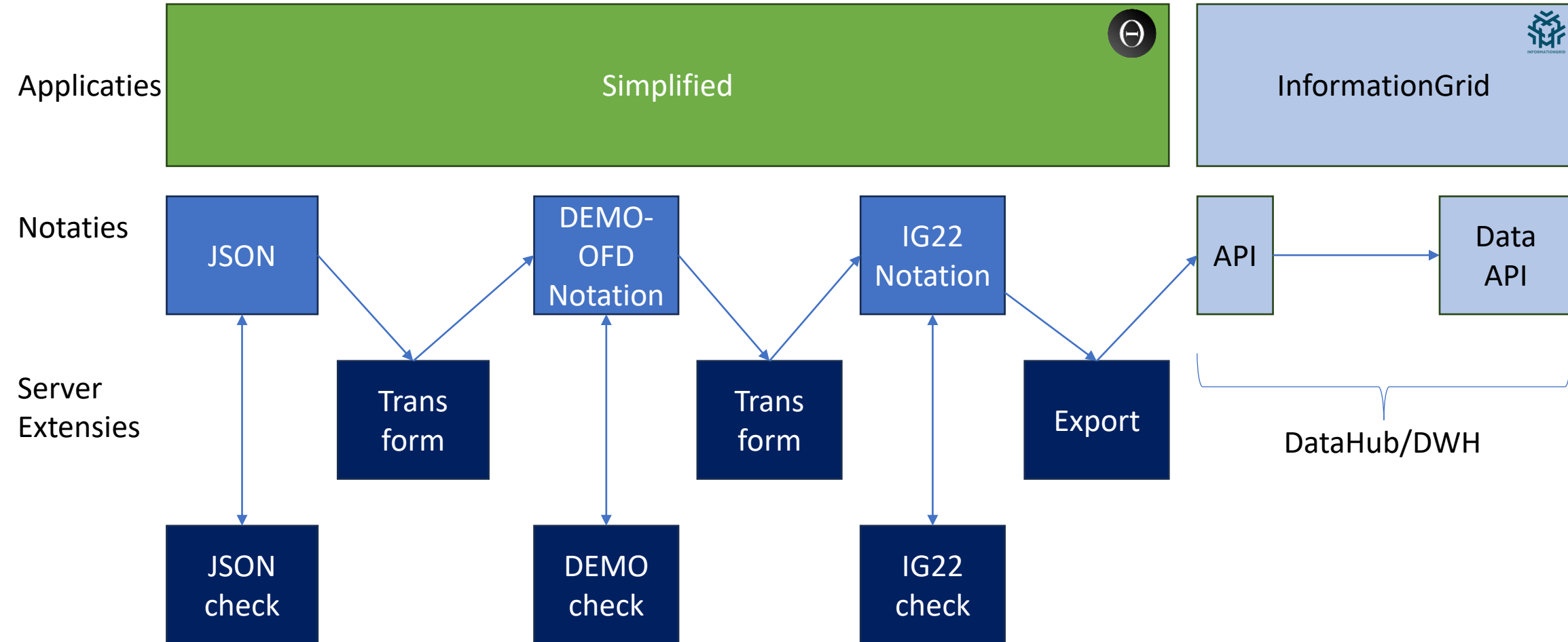
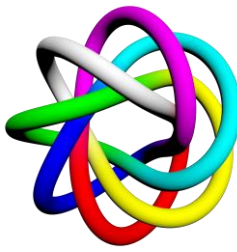


```
{
  "transactionroles": [
    {"id": "01", "transactionkindname": "registration starting", "actorrolename": "registration starter", "focus": "01"},
    {"id": "02", "transactionkindname": "registration paying", "actorrolename": "registration payer", "focus": "02"},
    {"id": "03", "transactionkindname": "registration ending", "actorrolename": "registration ender", "focus": "03"},
    {"id": "04", "transactionkindname": "registration management", "actorrolename": "registration manager", "focus": "04"}
  ],
  "compositeactorroles": [
    {"id": "01", "name": "(aspirant) member"}
  ],
  "multipletransactionkinds": [
    {"id": "91", "name": "Costs and terms"},
    {"id": "92", "name": "Person & living facts"}
  ],
  "initiates": [
    {"ctar": "01", "tk": "01"},
    {"ear": "01", "tk": "02"},
    {"ear": "04", "tk": "04"},
    {"ear": "04", "tk": "02"},
    {"ear": "04", "tk": "03"},
    {"ctar": "01", "tk": "03"}
  ]
}
```



Current State – Staging, CRUD

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Example

```
{
  "id": "string",
  "version": 0,
  "UBW_Grootboek": {
    "recordId": "string",
    "Periode": "string",
    "Bedrag": 0,
    "Waarde": "string",
    "Uren": 0,
    "Aantal": 0,
    "Tekst": "string",
  }
}
```

▼ Grootboek

- UBW_Grootboek
- UBW_Grootboek
- UBW_Grootboek
- UBW_Grootboek
- UBW_GrootboekAR
- UBW_GrootboekAR
- UBW_GrootboekDelta
- UBW_GrootboekDelta
- UBW_GrootboekLog
- UBW_GrootboekLogContent

CRUD

UBW_Grootboek
Aggregate Type

Journal id	SRO_BI_HAARLEM_Journal
Aggregate model	UBW_GrootboekAR
Aggregate root version property	version
Tenant aware	false
Create default projection	true

▼ Event type configurations

Event type identifier	Event data resource type
Added	UBW_Grootboek
Updated	UBW_Grootboek
Deleted	UBW_Grootboek

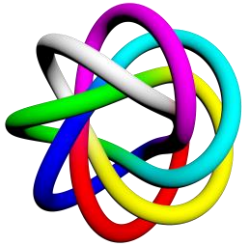
▼ Command type configurations

Command type identifier	Command label
Add	Add UBW Grootboek
Update	Update UBW Grootboek
Delete	Delete UBW Grootboek

UBW_Grootboek
Resource Type

State draft Basic DSL

Property	Type	Constraints
Record id [recordId]	string	Identifying Required
Periode [Periode]	string	
Bedrag [Bedrag]	float	
Waarde [Waarde]	string	
Uren [Uren]	double	
Aantal [Aantal]	float	
Tekst [Tekst]	string	



Future

